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#

PRODUCT: EV Battery Cell-to-Cell Interconnect (Cell Contact System)
REGIONS: United States, China, India, Europe
MATERIALS: Cu-dominant and Al-dominant designs per region
#

— 1. GLOBAL DATA

```
global_data = {
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "electrical_performance",
  "attribute_4": "mechanical_integrity",
  "attribute_5": "thermal_performance",    # STATE-DEPENDENT: baseline nominal
                                          # value stored per product; true value
                                          # varies by operating state (fast charge,
                                          # regen braking, cruise, cold start)

  # DIRECTIONS — "positive" = more is better; "negative" = less is better
  "direction_attribute_1": "negative",    # Lower cost is better
  "direction_attribute_2": "positive",    # Higher material continuity is better
                                          # (less disruption, lower switching risk)
  "direction_attribute_3": "positive",    # Higher electrical performance is better
                                          # (lower resistance, higher conductivity)
  "direction_attribute_4": "positive",    # Higher mechanical integrity is better
                                          # (stronger welds, better fatigue life)
  "direction_attribute_5": "positive",    # Higher thermal performance is better
                                          # (lower junction ΔT, better heat spreading)

  # ATTRIBUTE BOUNDS (global across all 8 products)
  # attribute_1: total_cost ($/vehicle)
  # MIN = India Al → 0.25×10.10 + 1.05×2.55 + 17.00 = $22.20
  # MAX = Europe Cu → 2.20×9.55 + 0.15×2.65 + 42.00 = $63.41
  "attribute_1_min": 22.20,
  "attribute_1_max": 63.41,
```

```

# attribute_2: material_continuity (normalized 0–1 scale)
# 0.00 = zero supply chain heritage; complete material switch
# 1.00 = fully entrenched; switching requires prohibitive reinvestment
"attribute_2_min": 0.00,
"attribute_2_max": 1.00,

# attribute_3: electrical_performance (normalized 0–1 scale)
# 0.52 = low-grade Al strip, poor weld contact, high joint resistance
# 1.00 = premium Cu, optimal weld geometry, minimal contact resistance
"attribute_3_min": 0.52,
"attribute_3_max": 1.00,

# attribute_4: mechanical_integrity (normalized 0–1 scale)
# 0.45 = poor weld quality, brittle joints, high vibration fatigue risk
# 1.00 = excellent weld integrity, proven fatigue life, full OEM qualification
"attribute_4_min": 0.45,
"attribute_4_max": 1.00,

# attribute_5: thermal_performance (normalized 0–1 scale; state-dependent)
# 0.48 = Al with poor thermal contact, high  $\Delta T$  at weld junction
# 1.00 = Cu with optimized thermal path, minimal junction  $\Delta T$ 
"attribute_5_min": 0.48,
"attribute_5_max": 1.00,
}

```

— 2. REGIONAL DATA

```

regional_data = [
{
  "region": "United States",
  "copper_price_per_kg": 9.50,    # COMEX HG basis + Midwest premium
                                # + C11000 ETP precision strip form premium
  "aluminum_price_per_kg": 2.80,  # LME + Section 232 tariff impact
                                # + Midwest Transaction Premium (~$0.30/kg)
                                # + precision strip form premium
  "copper_product_market_share": 0.55,
  "aluminum_product_market_share": 0.45,
  # Cost important but not overriding; performance competes strongly;
  # 800V architecture rollout elevates electrical priority;
  # moderate material inertia — OEM platform shifts are slow
  "weight_attribute_1": 0.30,
  "weight_attribute_2": 0.20,

```

```

    "weight_attribute_3": 0.25,
    "weight_attribute_4": 0.15,
    "weight_attribute_5": 0.10,      # Sum = 1.00 ✓
  },
  {
    "region": "China",
    "copper_price_per_kg": 9.80,      # SHFE basis + 13% import VAT floor
                                     # + warehouse/delivery premiums; offset
                                     # partly by OEM direct supply contracts
                                     # (Jiangxi Copper, Tongling Nonferrous)
    "aluminum_price_per_kg": 2.15,    # World's lowest Al price; ~57% of global
                                     # primary Al produced domestically;
                                     # Yunnan/Xinjiang smelter direct supply;
                                     # SHFE basis + domestic rolling scale discount
    "copper_product_market_share": 0.38,
    "aluminum_product_market_share": 0.62,
    # Most aggressive $/kWh targets globally — cost dominates all other attributes;
    # adequate-threshold mentality on performance; lowest material switching
    # resistance of all four regions; LFP thermal context elevates thermal weight
    "weight_attribute_1": 0.40,
    "weight_attribute_2": 0.15,
    "weight_attribute_3": 0.20,
    "weight_attribute_4": 0.13,
    "weight_attribute_5": 0.12,      # Sum = 1.00 ✓
  },
  {
    "region": "India",
    "copper_price_per_kg": 10.10,     # Highest Cu price of all 4 regions;
                                     # LME + 5% BCD + 0.5% SWS + CIF freight
                                     # to Indian ports + inland logistics;
                                     # Sterlite/Thoothukudi closure (2018)
                                     # created domestic refinery gap — India
                                     # imports >80% of refined copper needs
    "aluminum_price_per_kg": 2.55,    # Partial domestic supply (Hindalco/BALCO/NALCO);
                                     # EV-grade precision strip requires import
                                     # premium; 7.5% BCD; coal-based smelting
                                     # energy cost moderate vs European
    "copper_product_market_share": 0.33,
    "aluminum_product_market_share": 0.67,
    # Extreme price sensitivity dominates all other considerations;
    # lowest material continuity weight — youngest EV market, most open to
    # material optimization; elevated mechanical weight for India road roughness
    # (high IRI values); elevated thermal weight for 35–45°C ambient temps
    "weight_attribute_1": 0.45,

```

```

    "weight_attribute_2": 0.12,
    "weight_attribute_3": 0.13,
    "weight_attribute_4": 0.18,
    "weight_attribute_5": 0.12,      # Sum = 1.00 ✓
  },
  {
    "region": "Europe",
    "copper_price_per_kg": 9.55,      # LME Grade A basis + Rotterdam physical
                                     # premium (~$80/ton) + precision strip
                                     # form premium; 0% EU MFN import duty
                                     # on refined copper (critical material)
    "aluminum_price_per_kg": 2.65,    # LME + elevated EU physical premium
                                     # (~$300/ton); post-2022 energy crisis
                                     # smelter curtailments; industrial
                                     # electricity €0.12–0.18/kWh vs $0.04–0.06
                                     # in Chinese hydro regions; CBAM provisioning
                                     # from 2026 upward-pressuring import pricing
    "copper_product_market_share": 0.53,
    "aluminum_product_market_share": 0.47,
    # Premium OEM culture — cost less dominant than other regions;
    # material continuity and electrical performance carry equal highest weight
    # reflecting: 3–5yr OEM validation cycles (IATF 16949 + DV/PV) and
    # 800V platform prevalence (BMW Neue Klasse, Porsche PPE, Mercedes MMA)
    "weight_attribute_1": 0.25,
    "weight_attribute_2": 0.25,
    "weight_attribute_3": 0.25,
    "weight_attribute_4": 0.15,
    "weight_attribute_5": 0.10,      # Sum = 1.00 ✓
  },
]

```

— 3. PRODUCT DATA

```

# 8 entries: Cu-dominant and Al-dominant for each of the 4 regions
#
# Cu-dominant design (per vehicle, ~75kWh cylindrical 21700 pack, ~4,400 cells):
#  copper_kg  = 2.20 (main CCS strips, Ni-plated C11000, laser-welded to cell caps)
#  aluminum_kg = 0.15 (trace Al at some positive terminal interfaces)
#  total_mass = 2.35 kg
#
# Al-dominant design (sized for equivalent current capacity):
#  copper_kg  = 0.25 (retained at negative terminal bimetallic contact zones;
#  direct Al-to-Cu-cap welding requires Cu/Al transition strips)

```

aluminum_kg = 1.05 (main CCS strips; 1.64× cross-section area vs Cu at 0.49× mass)
 # total_mass = 1.30 kg (~45% mass reduction vs Cu-dominant)
 #
 # attribute_5 (thermal_performance): STATE-DEPENDENT
 # Values below are BASELINE NOMINAL at ~1C discharge rate
 # True values vary dynamically: lower during DC fast charge / regen peaks,
 # higher during low-rate cruise; India baseline uses 35°C ambient reference

product_data = [

```
# [
#   UNITED STATES — Cu   |
# ]
{
  "region": "United States",
  "dominant material": "cu",
  "product": "EV Battery Cell-to-Cell Interconnect",
  "copper_kg": 2.20,
  "aluminum_kg": 0.15,
  "total_mass_kg": 2.35,
  "non_material_cost_per_unit": 35.00,
  # total_cost = 2.20×9.50 + 0.15×2.80 + 35.00 = $56.32

  "attribute_2_value": 0.82,
  # Strong Cu heritage in US EV sector (Tesla cylindrical, Ford/LGES,
  # Rivian); growing prismatic pressure from GM Ultium creates some
  # switching risk; incumbent Cu laser weld supply chains deeply embedded;
  # full re-tooling + process re-qualification required to switch to Al

  "attribute_3_value": 0.87,
  # Cu ~100% IACS leveraged strongly; US 800V platforms (Tesla 4680,
  # Rivian) demand tight resistance budgets — Cu meets comfortably;
  # slight discount for weld joint contact resistance variability
  # in high-volume cylindrical cell production

  "attribute_4_value": 0.83,
  # Accordion-relief geometry handles cylindrical cell breathing and
  # automotive vibration loads well; IATF 16949 compliance standard
  # across US Tier-1 suppliers; laser weld process control mature

  "attribute_5_value": 0.82,
  # Baseline nominal (~25°C ambient, ~1C rate)
  # Cu thermal conductivity ~390 W/m·K; effective heat spreading
  # from cell caps to pack cooling structure; adequate headroom
```

```

    # before cell cap temperature limits reached under fast charge
  },

  # [ UNITED STATES — Al | ]
  # [ ]
  {
    "region": "United States",
    "dominant material": "al",
    "product": "EV Battery Cell-to-Cell Interconnect",
    "copper_kg": 0.25,
    "aluminum_kg": 1.05,
    "total_mass_kg": 1.30,
    "non_material_cost_per_unit": 30.00,
    # total_cost = 0.25×9.50 + 1.05×2.80 + 30.00 = $35.32

    "attribute_2_value": 0.52,
    # Weakest Al material lock-in of all four regions; US cell-level
    # interconnects historically Cu-dominant; Al used in some prismatic
    # formats (GM Ultium) but not deeply embedded; risk of reversion
    # to Cu grows as 800V cylindrical platforms scale up

    "attribute_3_value": 0.65,
    # Al ~61% IACS vs Cu 100%; larger cross-section compensates volume
    # but joint resistance still penalized; US 800V platforms impose
    # tightest resistance budgets — Al is a meaningful compromise;
    # 2nd lowest electrical score among Al-dominant designs

    "attribute_4_value": 0.72,
    # Adequate for prismatic formats; Al laser welding more challenging
    # (high reflectivity, oxide layer, hydrogen porosity risk);
    # fatigue life meets IATF standards but with lower inherent margin
    # than Cu; accordion-relief geometry still required

    "attribute_5_value": 0.62,
    # Baseline nominal (~25°C ambient, ~1C rate)
    # Al thermal conductivity ~205 W/m·K (~47% of Cu);
    # adequate at moderate conditions; reduced headroom
    # under DC fast charge thermal loading vs Cu design
  },

  # [ CHINA — Cu | ]
  # [ ]
  # [ ]

```

```

{
  "region": "China",
  "dominant material": "cu",
  "product": "EV Battery Cell-to-Cell Interconnect",
  "copper_kg": 2.20,
  "aluminum_kg": 0.15,
  "total_mass_kg": 2.35,
  "non_material_cost_per_unit": 22.00,
  # total_cost = 2.20×9.80 + 0.15×2.15 + 22.00 = $43.88

  "attribute_2_value": 0.62,
  # Cu used in CATL cylindrical lines (export + domestic NMC); however
  # majority of China's market has migrated to Al for prismatic LFP;
  # Cu continuity exists but under structural market pressure; lower
  # inertia than US/EU — Chinese OEMs have demonstrated willingness
  # to switch materials at scale

  "attribute_3_value": 0.86,
  # Strong manufacturing capability (CATL, BYD); slight discount vs EU
  # reflecting quality variation across broad tier-2/3 supplier base;
  # 400V dominant architecture reduces 800V resistance criticality vs
  # US/EU but Cu still clearly outperforms Al in this context

  "attribute_4_value": 0.80,
  # Good quality from leading Tier-1 suppliers; CATL/BYD internal
  # standards high; some variation across China's diverse tier-2/3
  # supplier ecosystem; GB/T standards less stringent than ECE/IATF

  "attribute_5_value": 0.80,
  # Baseline nominal (~25°C ambient, ~1C rate)
  # Cu thermal conductivity strong; LFP chemistry generates less heat
  # per cell than NMC — lower thermal flux through interconnect;
  # good thermal performance in typical Chinese LFP operating context
},

```

```

# [ CHINA — Al | ]
# [ ]
# [ ]
{
  "region": "China",
  "dominant material": "al",
  "product": "EV Battery Cell-to-Cell Interconnect",
  "copper_kg": 0.25,
  "aluminum_kg": 1.05,

```

"total_mass_kg": 1.30,
"non_material_cost_per_unit": 18.00,
total_cost = 0.25×9.80 + 1.05×2.15 + 18.00 = \$22.71

"attribute_2_value": 0.82,
Strongest Al heritage of all four regions; CATL prismatic NMC/LFP
and BYD Blade Battery have built entire interconnect supply chains
around Al; stamping tooling, surface treatment, laser weld parameter
libraries, and QC all optimized for Al; highest Al lock-in globally

"attribute_3_value": 0.68,
Al conductivity acceptable in LFP prismatic context: larger cell format
(~100–200 large cells vs 4,400 cylindrical) means resistance managed
via cross-section sizing; 400V dominant architecture (vs 800V) relaxes
resistance budget vs US/EU; 2nd best electrical score among Al designs

"attribute_4_value": 0.75,
Best mechanical integrity among Al-dominant designs globally;
CATL/BYD have refined Al weld processes at extreme volume — weld
parameter optimization, post-weld inspection, accelerated fatigue
testing all mature; prismatic cell format well-characterized for
Al interconnect swelling accommodation geometry

"attribute_5_value": 0.64,
Baseline nominal (~25°C ambient, ~1C rate)
Best thermal score among Al-dominant designs; LFP chemistry generates
less heat than NMC — lower thermal flux partially compensates for
Al's lower thermal conductivity (~205 W/m·K vs Cu ~390 W/m·K)

},

#

INDIA — Cu	
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#

{

"region": "India",
"dominant material": "cu",
"product": "EV Battery Cell-to-Cell Interconnect",
"copper_kg": 2.20,
"aluminum_kg": 0.15,
"total_mass_kg": 2.35,
"non_material_cost_per_unit": 20.00,
total_cost = 2.20×10.10 + 0.15×2.55 + 20.00 = \$42.60

"attribute_2_value": 0.55,

Lowest Cu material continuity of all four regions; India's EV market
young (mass adoption post-2021) — neither material deeply entrenched;
economic forces have favored AI from market inception, meaning Cu
designs face headwinds and weaker supply chain lock-in than US/EU

"attribute_3_value": 0.81,

Cu conductivity advantage present but discounted vs US/EU: India's
manufacturing infrastructure still maturing; 2/3-wheeler dominant
market operates at lower currents (5–20 kW peak) vs passenger EV
(100–300 kW) — full Cu conductivity benefit less critical here

"attribute_4_value": 0.74,

Lowest mechanical integrity for Cu-dominant designs; India's road
roughness (high IRI) creates most severe vibration fatigue loading
globally; even Cu's superior fatigue properties are stressed by
the harsh load spectrum; less mature QC ecosystem adds weld risk

"attribute_5_value": 0.78,

Baseline nominal (~35°C ambient India reference, ~1C rate)
Cu thermal conductivity strong (~390 W/m·K) but India's elevated
ambient temperature (35–45°C) structurally reduces available thermal
headroom before cell cap temperature limits reached; lowest thermal
score among Cu-dominant designs across all four regions

},

#

INDIA — AI	
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{

"region": "India",
"dominant material": "al",
"product": "EV Battery Cell-to-Cell Interconnect",
"copper_kg": 0.25,
"aluminum_kg": 1.05,
"total_mass_kg": 1.30,
"non_material_cost_per_unit": 17.00,
total_cost = 0.25×10.10 + 1.05×2.55 + 17.00 = \$22.20
← GLOBAL MINIMUM total cost across all 8 products

"attribute_2_value": 0.62,

Growing AI inertia — cost economics have driven AI adoption from market
inception; Ola Electric, Ather Energy, TVS building AI-oriented supply
chains; inertia building but not yet as structurally embedded as China;
market young enough that switching is conceivable but becoming less likely

```

"attribute_3_value": 0.70,
# Best electrical performance score among Al-dominant designs; India's
# 2/3-wheeler applications (5–20 kW peak) impose the least demanding
# current loads of all regional applications — Al's conductivity deficit
# (~61% IACS) is least penalizing at these power levels; adequate
# performance within the 2/3-wheeler operating envelope

"attribute_4_value": 0.66,
# GLOBAL MINIMUM mechanical integrity across all 8 products;
# most critical structural concern in the dataset: India's severe road
# roughness × Al's lower fatigue strength at weld joints = highest
# structural failure risk of any region-material combination; less
# mature QC ecosystem compounds weld consistency risk

"attribute_5_value": 0.60,
# Baseline nominal (~35°C ambient India reference, ~1C rate)
# GLOBAL MINIMUM thermal performance across all 8 products;
# compound of two adverse factors: (1) Al thermal conductivity
# ~205 W/m·K; (2) India's 35–45°C ambient — smallest thermal
# headroom of any region-material combination; partially offset
# by lower 2/3-wheeler current magnitudes
},

#
# | EUROPE — Cu |
#
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "EV Battery Cell-to-Cell Interconnect",
  "copper_kg": 2.20,
  "aluminum_kg": 0.15,
  "total_mass_kg": 2.35,
  "non_material_cost_per_unit": 42.00,
  # total_cost = 2.20×9.55 + 0.15×2.65 + 42.00 = $63.41
  # ← GLOBAL MAXIMUM total cost across all 8 products

  "attribute_2_value": 0.88,
  # GLOBAL MAXIMUM material continuity across all 8 products;
  # EU OEM validation cycles 3–5 years (IATF 16949 + OEM-specific DV/PV);
  # any material change triggers full re-validation; Tier-1 suppliers
  # (Manz, Grob, Samsung SDI packs) have deeply optimized Cu weld
  # parameters; highest switching cost of any region-material combination

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```

"attribute_3_value": 0.90,
# GLOBAL MAXIMUM electrical performance across all 8 products;
# 800V architecture standard on European premium platforms (BMW Neue
# Klasse, Porsche PPE/J1, Mercedes MMA, Audi PPE) — world's tightest
# interconnect resistance budgets; Cu conductivity advantage most fully
# leveraged; EU Tier-1 manufacturing quality highest globally

"attribute_4_value": 0.87,
# GLOBAL MAXIMUM mechanical integrity across all 8 products;
# ECE regulations + OEM-specific fatigue protocols (BMW GS 95024,
# VW PV 1200) most stringent globally; heavy investment in SPC,
# in-line optical + X-ray weld inspection; cool EU climate reduces
# thermal fatigue cycling amplitude vs warmer regions

"attribute_5_value": 0.84,
# Baseline nominal (~15°C ambient EU reference, ~1C rate)
# GLOBAL MAXIMUM thermal performance across all 8 products;
# three compounding advantages: (1) Cu ~390 W/m·K conductivity;
# (2) EU manufacturing quality minimizes contact thermal resistance;
# (3) cool European ambient (~10–20°C avg) maximizes thermal headroom
# — largest delta between operating temperature and ambient limit
},

#
# | EUROPE — Al |
#
{
  "region": "Europe",
  "dominant material": "al",
  "product": "EV Battery Cell-to-Cell Interconnect",
  "copper_kg": 0.25,
  "aluminum_kg": 1.05,
  "total_mass_kg": 1.30,
  "non_material_cost_per_unit": 36.00,
  # total_cost = 0.25×9.55 + 1.05×2.65 + 36.00 = $41.17

  "attribute_2_value": 0.58,
  # Moderate Al heritage; Al used in some mid-tier prismatic applications
  # (Renault Zoe, Stellantis e-CMP) but EU premium segment is Cu-dominant;
  # market trend toward cylindrical/800V favors Cu; Al inertia exists in
  # mid-tier but faces structural headwinds; lower lock-in than China
  # because EU OEM mix heavily skews toward premium platforms

```

"attribute_3_value": 0.63,
GLOBAL MINIMUM electrical performance across all 8 products;
most adverse combination in dataset: Al conductivity deficit (~61% IACS)
× EU's tightest 800V resistance budgets (Porsche, BMW, Mercedes);
even in non-800V EU platforms, OEM internal resistance specs exceed
all other regions' requirements; Al is the most significant compromise here

"attribute_4_value": 0.72,
Adequate mechanical integrity; EU Tier-1 suppliers invest heavily
in Al laser weld process control (Constellium/Novelis qualified
strip stock); ECE fatigue requirements met with optimized Al relief
geometry; cool EU climate reduces thermal fatigue cycling vs India;
matches US Al score — same inherent material property baseline

"attribute_5_value": 0.61,
Baseline nominal (~15°C ambient EU reference, ~1C rate)
Al thermal conductivity limited (~205 W/m·K); partially offset by
cool EU ambient maximizing thermal headroom; partially negated by
800V architecture's higher power density increasing thermal flux
through interconnects; net result: 2nd lowest thermal score
among Al-dominant designs (above India Al at 0.60)

},

]